

B Suraj Patra

B.Tech CSE Student | Software Engineering & Full-Stack Development | Open-Source Contributor
2300031514cseelge@gmail.com | +91 8763232589 | Vijayawada, India
bsurajpatra.me | linkedin.com/in/b-suraj-patra | github.com/bsurajpatra

Summary

A Computer Science undergraduate with hands-on involvement in large-scale engineering projects and production-grade full-stack systems. Demonstrated ownership through open-source contributions and real-world deployments, with a strong foundation in data structures and algorithms.

Skills

Technical Skills

- Programming: C, Java, JavaScript, SQL
- CS Fundamentals: Data Structures, Algorithms, Object-Oriented Programming
- Full-Stack: MERN (MongoDB, Express.js, React, Node.js), REST APIs, React Native
- DevOps & Tools: Git, GitHub, GitHub Actions, Docker

Soft Skills

- Problem Solving, Critical Thinking, Leadership, Collaboration, Agile/Scrum

Projects

KL Eats

- Architected and scaled a campus food-ordering platform serving 4.9K+ users with 1500+ transactions, achieving 99.9% uptime and sustaining peak load of 63+ RPS (8650 requests via Locust). Improved frontend performance score 59→86 and reduced First Contentful Paint 1.7s→0.6s by optimizing Core Web Vitals through caching, UI restructuring, and Redis-backed session management. Built automated finance reconciliation and Telegram-based order orchestration enabling Rs.1.5L+ monthly transaction processing; operating under Equitech Lab Pvt. Ltd.
- **Tech Stack:** Next.js 15, React 18, TypeScript, Tailwind, shadcn/ui, Node.js, Express.js, MySQL, Redis

FaceAttend

- Architected a privacy-first facial recognition attendance system with a distributed architecture spanning mobile kiosk capture, real-time backend orchestration, ERP management portal, and a dedicated FaceNet microservice. Implemented embedding-only biometric storage with cosine similarity matching, multi-embedding rotation (up to 5 per user), and duplicate-face prevention, achieving 95% operational accuracy with 1-2s recognition latency. Reduced inference calls by 60% via on-device face pre-filtering and enforced device trust, audit logging, rate-limited authentication, and Android Lock Task Mode; developed with Creovsys Solutions India Pvt. Ltd.
- **Tech Stack:** React Native (Expo), React (Vite), TypeScript, Node.js, Express.js, MongoDB, Python, Flask, FaceNet (facenet-pytorch), Socket.IO, JWT

Education

K L E F Deemed To Be University – B.Tech CSE (2023—2027) — GPA: 9.40/10 — Vijayawada, India
Sunabeda Public School – Class XII (ISC, 2021—2023) — GPA: 8.50/10 — Sunabeda, Odisha, India
Sunabeda Public School – Class X (ICSE, 2020—2021) — GPA: 9.35/10 — Sunabeda, Odisha, India

Certifications

MongoDB Associate Developer | AWS Cloud Practitioner | GitHub Foundations

Patents & Publications

- **Biometric-based Voting System with Multi-layer Authentication** — IN202541073801 (Aug 2025). Invented a multi layer biometric voter authentication system to prevent impersonation and improve election transparency.
- **DigiChit : Automated Chit Fund Processing System** — IN202541098116 (Nov 2025). Developed a chit fund processing system integrating rule-based mandate verification, fraud-detection logic, and compliance automation.

Achievements

- Competitive Programming: LeetCode (Rating: 1600+) | CodeChef (Rating: 1500+)
- **Hackathons & Competitions:** Finalist, Build for Bharat Hackathon; Semi-finalist, Code for Bharat Season 2; Top 15, CodeRush Competitive Programming Challenge (IIITM Gwalior) — Team Code4Cause.
- Contributed to 9+ open-source repositories and supported open-data initiatives across Wikipedia, OpenStreetMap, and Weblate for regional-language accessibility.

Leadership

- **Head, Wikipedia Club** — **GNU/Linux Users Group (GLUG):** Leading open-data and Wikipedia contribution initiatives, mentoring students, and promoting open knowledge through structured community participation.